

Efficient Algorithms for Lasso Regression Report

Yuehangsha Huang Mengying Li
Applied Machine Learning in Python – LMU
Yuehangsha.Huang@campus.lmu.de Li.Mengying@campus.lmu.de

July 2, 2026

1 Task Overview

This project studies efficient algorithms for the Lasso regression problem

$$\min_w \frac{1}{2n} \|Xw - y\|_2^2 + \lambda \|w\|_1.$$

The Lasso objective is convex but non-smooth due to the ℓ_1 penalty, which promotes sparsity by shrinking many coefficients to zero; we tested this on synthetic data with $n = 100$, $d = 200$, Gaussian noise $\sigma = 0.5$, and a true coefficient vector with 10 nonzero entries.

We evaluated Subgradient Descent, ISTA, FISTA, and Coordinate Descent in terms of convergence speed, final sparsity, Lasso path behavior, and robustness to initialization and regularization strength, using sklearn’s Lasso and the Diabetes dataset as reference validations.

2 Methods

Subgradient Descent was used as the baseline. ISTA applies the proximal update

$$w^{(k+1)} = S_{\eta\lambda} \left(w^{(k)} - \eta \nabla g(w^{(k)}) \right),$$

where $S_\alpha(z) = \text{sign}(z) \max(|z| - \alpha, 0)$ is the soft-thresholding operator. FISTA accelerates ISTA, and Coordinate Descent updates one coefficient at a time using the same soft-thresholding idea.

3 Experiments and Results

3.1 Convergence

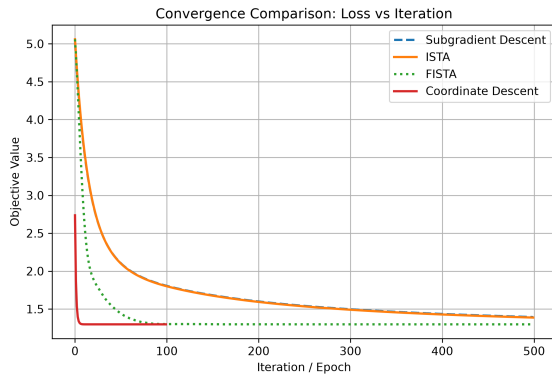


Figure 1: Objective value versus iteration.

Figure 1 compares the convergence of the four solvers. Subgradient Descent and ISTA decrease slowly and their curves almost overlap under the chosen step size. FISTA converges much faster because of Nesterov acceleration. Coordinate Descent reaches the lowest objective within only a few epochs. The final objective values are 1.393, 1.387, 1.298 and 1.298 for Subgradient, ISTA, FISTA and Coordinate Descent, respectively.

3.2 Sparsity and Lasso Path

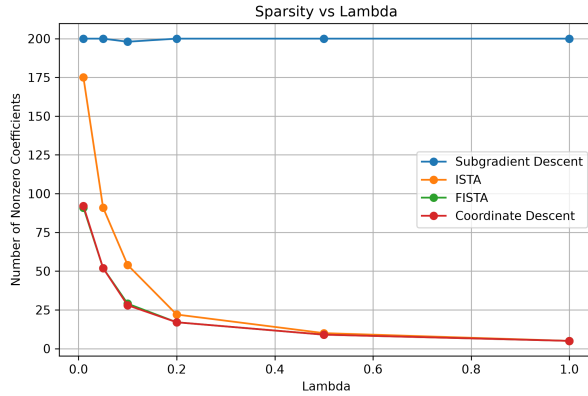
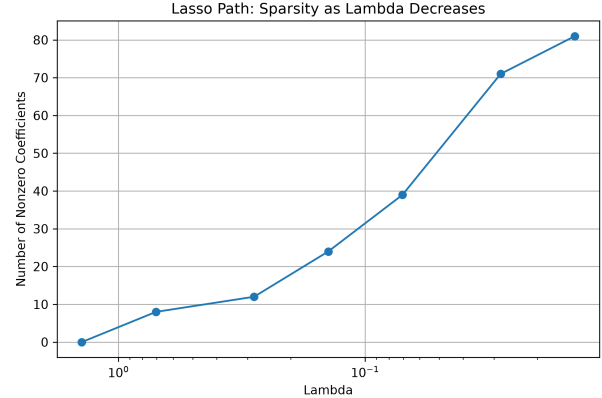
Figure 2: Sparsity versus λ .Figure 3: Lasso path as λ decreases.

Figure 2 shows that increasing λ leads to fewer nonzero coefficients. Subgradient Descent remains dense because it rarely sets coefficients exactly to zero, while ISTA, FISTA, and Coordinate Descent produce much sparser solutions through soft-thresholding. Figure 3 confirms the Lasso path behavior: as λ decreases, regularization weakens and more coefficients enter the model.

3.3 Sparse Recovery

Table 1: Recovery error.

Method	$\ \hat{w} - w^*\ _2$
Subgradient	1.486
ISTA	1.483
FISTA	0.748
Coordinate	0.749

Table 2: Support recovery.

Method	Prec.	Rec.	F1
Subgradient	0.051	1.000	0.096
ISTA	0.185	1.000	0.313
FISTA	0.345	1.000	0.513
Coordinate	0.357	1.000	0.526

Tables 1 and 2 show that all methods recovered the 10 true nonzero coefficients, so recall equals 1. However, Subgradient Descent selected many false positives and had very low precision. FISTA and Coordinate Descent achieved much smaller recovery errors and higher precision, showing better sparse recovery.

Our Coordinate Descent implementation matched `sklearn` Lasso almost exactly, with the same objective value (1.2976), 28 nonzero coefficients, and recovery error 0.7485.

3.4 Robustness

3.4.1 Robustness to Initialization

The initialization experiment shows that all four solvers are robust to the choice of initial coefficients. With zero, small random, and large random initialization, they reached nearly the same final objective after sufficient iterations, indicating that initialization has little effect in the long-run setting.

It is worth noting that under a limited iteration budget, solver differences become more apparent. FISTA and Coordinate Descent reach lower objective values much faster, while Subgradient Descent and ISTA converge more slowly.

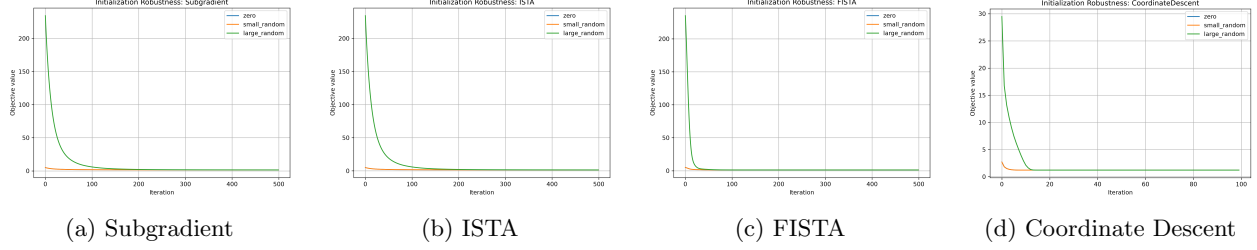
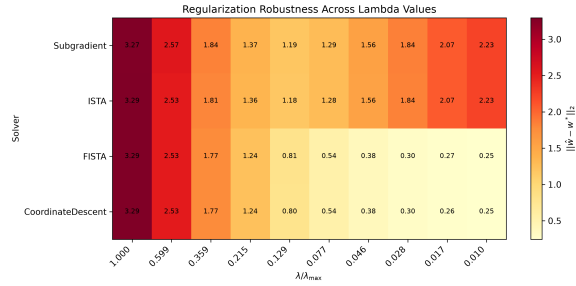


Figure 4: Objective value vs. iterations under different initialization schemes for each solver.

3.4.2 Robustness to Regularization Strength

We varied $\lambda/\lambda_{\max}$ and measured the recovery error $\|\hat{w} - w^*\|_2$, which is available only for synthetic data with known ground-truth coefficients.

Figure 5: Heatmap of distance to true coefficients across λ values.

The heatmap shows that ISTA, FISTA, and Coordinate Descent behave almost identically, with recovery error steadily decreasing as λ becomes smaller. In contrast, Subgradient Descent is more sensitive to λ and performs best only at an intermediate regularization level.

Overall, proximal and coordinate-wise methods are more robust to regularization strength and recover the sparse ground-truth vector more accurately than Subgradient Descent.

3.5 Validation on the Diabetes Dataset

We further validated the results on the Diabetes dataset using test MSE, sparsity, and selected features. FISTA, Coordinate Descent, and sklearn Lasso achieved nearly identical performance and selected the same seven features, while ISTA chose eight features and Subgradient Descent gave a denser solution with all ten features selected.

Solver	Best λ ratio	Best test MSE	Sparsity at best λ	Selected features
Subgradient	0.046416	2801.133037	10	bmi, s5, bp, s3, sex, s1, s6, s2, s4, age
ISTA	0.046416	2801.078318	8	bmi, s5, bp, s3, sex, s1, s6, s2
FISTA	0.046416	2798.609930	7	bmi, s5, bp, s3, sex, s1, s6
Coordinate Descent	0.046416	2798.601225	7	bmi, s5, bp, s3, sex, s1, s6
sklearn Lasso	0.046416	2798.601223	7	bmi, s5, bp, s3, sex, s1, s6

Table 3: Best performance and selected features on the Diabetes dataset.

4 Discussion

Soft-thresholding methods are more effective for Lasso optimization than Subgradient Descent. Coordinate Descent performed best overall, while FISTA also converged quickly and recovered sparse solutions accurately. Subgradient Descent was useful as a baseline but produced less sparse solutions with more false positives.